Internship Opportunity – Towards a Self-Growing Training Dataset: Automatic Extraction and Selection of Defect Samples from DAS Signals

Background

Distributed Acoustic Sensing (DAS) turns an optical fibre into a continuous array of acoustic sensors capable of measuring vibrations over several kilometres. Its operation relies on Rayleigh backscattering: when a light pulse travels through the fibre, a small fraction of the light is scattered back towards the source due to microscopic irregularities in the glass. The phase and amplitude of the backscattered light change when the fibre is strained, making it possible to detect and locate vibrational events.
The SURFO team uses DAS to monitor railway infrastructure in real time. One of the most safety-critical applications is the detection of track defects that produce localised vibration signatures when trains pass over them. A major challenge is the extreme scarcity of confirmed examples of the target phenomena in the available data, preventing direct training or validation of a model on the phenomenon of interest.
To overcome this limitation, the team uses a proxy-based strategy: rather than searching directly for a target phenomenon that is absent from the data, it studies a physically related track defect that is more frequent and observable at the test site. An initial signal-processing pipeline has been developed to extract spatiotemporal windows around calibrated reference positions for each passing train, then characterise and classify the corresponding signature. This pipeline provides quantitative preselection, but each candidate must then be visually reviewed by an expert using time–distance representations (*waterfalls*) before being added to the reference dataset.

Challenge

This visual review is currently the main bottleneck. It is time-consuming, difficult to reproduce and severely limits the size of the training dataset available for the future classification module. Three challenges arise:

- **Very few positive examples**: only a few dozen validated instances are available from a small number of sites, creating a risk that a supervised model learns site identity rather than the defect signature.
- **Severe imbalance and no direct labels**: only the position of the track component is known; individual train passages are not inherently labelled as defective.
- **Ambiguous signatures**: normal train passages, civil-engineering structures, variations in rolling stock and train speed can produce confounding signatures.

However, there is a considerable amount of unlabelled data: every train passage at a recorded reference position is a potentially useful sample. Self-supervised and few-shot approaches are well suited to this setting.

Internship objective

Develop a semi-automatic pipeline for selecting samples from DAS signals, replacing systematic visual inspection with a reliable preselection model and enabling continuous, large-scale enrichment of the training dataset.

Internship tasks

The work will focus on the following topics:

- **Reference-position database and automated extraction**: structure the calibrated positions of monitored track components, then automatically extract and normalise time–distance patches for each detected train passage.
- **Representation learning**: explore self-supervised pretraining (masked autoencoders, contrastive learning) or transfer learning from pretrained vision models; use the few validated samples as prototypes.
- **Similarity scoring and classification**: combine similarity to prototypes with anomaly detection against normal train passages, and compare with a baseline based on the existing expert-designed features.
- **Active-learning loop**: propose informative candidates for expert review, incorporate successive validations and measure the reduction in annotation effort.

The first two topics will take priority; the other two may be pursued depending on progress. The internship will cover the existing processing pipeline, a review of few-shot, self-supervised and semi-supervised learning, algorithm design and evaluation, validation by site and date, and transferability to other instrumented sites.

Expected outcomes

- A usable reference-position database and an automated sample-extraction pipeline for individual train passages;
- A preselection model that reduces visual inspection, evaluated using precision at top ranks and the rate of missed defects;
- An enriched, documented and reusable reference dataset, together with an assessment of the limitations of the proxy-based strategy.

Candidate profile

Education

Final-year engineering student or second-year Master's student in signal processing, computer vision, artificial intelligence or applied mathematics.

Technical skills

- Strong foundations in signal processing, image processing and/or *machine learning*;
- Proficiency in Python (NumPy, SciPy, scikit-learn and PyTorch are appreciated);
- Interest in learning from limited data (few-shot learning, self-supervised learning and anomaly detection);
- Ability to analyse experimental data.

Personal qualities

- Scientific rigour, curiosity, independence and critical thinking;
- Teamwork and good communication skills.

Practical information

- Duration: 5 to 6 months
- Location: La Plaine-Saint-Denis, France
- Start date: from 1 March 2027
- Supervision: BEN AMARA Imen
- Allowance: in accordance with applicable regulations
- Benefits: SNCF travel benefits.

How to apply

Please send your CV and cover letter to all the following addresses:

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References

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